HOMEWORK 6.8 · UNIT 6 · NC.8.F.2 · NC.8.F.4 · NC.8.F.3

Linear Functions from Tables

Targets: F.2-1, F.4-4, F.3-1, F.3-4, F.2-2, F.4-6, F.3-6, F.4-5, F.3-9, F.2-6

Name _____ Date _____ Block _____

Remember: Slope from a table: $m = \frac{\text{change in } y}{\text{change in } x}$. ***b*** = the value of y when $x = 0$. Ex: x: 0, 1, 2 and y: 5, 8, 11. Each time x goes up 1, y goes up 3. So $m = 3$, $b = 5$, and $y = 3x + 5$. If the table skips $x = 0$, work backward to find b.

SCORE 3.0 Approaching · the basics

Targets: F.2-1, F.4-4, F.3-1, F.3-4, F.2-2

1 Find the rate of change and the initial value. F.2-1

x	0	1	2	3
y	2	8	14	20

2 Which equation fits the table? F.4-4

x	0	1	2	3
y	20	16	12	8

(A) $y = -4x + 20$
(B) $y = 4x + 20$
(C) $y = -4x + 8$
(D) $y = 20x - 4$

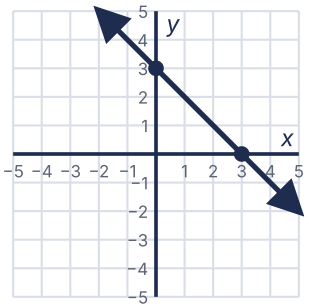
3 Is the table linear? REVIEW 6.4 F.3-1

x	1	2	3	4
y	3	6	10	15

Linear Not linear

Why? _____

4 Which equation matches the graph? REVIEW 6.7 F.3-4



(A) $y = -x + 3$
(B) $y = x + 3$
(C) $y = 3x - 1$
(D) $y = -x - 3$

5 In a table, the initial value is the value of y when $x = 0$. F.2-2

True False

SCORE 4.0 Proficient · on target

Targets: F.4-6, F.2-2, F.3-6, F.4-5

6 A tank is drained. Write an equation for the water y after x minutes. What does the slope mean? F.4-6,F.2-2

Minutes	0	5	10	15
Water (gal)	200	175	150	125

7 **Caleb** made a mistake. Circle it. Explain it. Fix it. F.3-6

x	1	2	3
y	7	10	13

CALEB'S WORK

$m = 3$

$b = 7$ (the first y-value)

$y = 3x + 7$

8 Find the slope of the line through (−2, 6) and (2, −2). REVIEW 6.5 F.4-5

SCORE 5.0 Mastery · go deeper

Targets: F.3-9, F.2-6

9 This table is linear. Write its equation. Explain how you found ***b***. F.3-9

x	2	5	11
y	1	7	19

m = _____ . To find b, I _____ . The equation is _____ .

10 This table is linear. Find the missing value k. Explain. F.2-6

x	2	4	10
y	9	15	k

The rate of change is _____ . So k = _____ because _____ .